A

Industrial-Oriented Mini

Project On

A DEEP LEARNING MODEL FOR BRAIN STROKE DETECTION USING CT SCANS

(Submitted in partial fulfilment of the requirements for the award of Degree)

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CERTIFICATE

This is to certify that the project entitled "**A DEEP LEARNING MODEL FOR BRAIN STROKE DETECTION USING CT SCANS**" being submitted by **S.Sathya Jaya Sree (237R1A05J1), N.Niharika (237R1A05G6) & H.RamaKanth(237R1A05E9)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

This project is titled as "*A deep learning model for brain stroke detection using CT scan.*" Stroke is a critical medical condition and one of the leading causes of death and long-term disability worldwide. Early and accurate detection is essential for effective treatment and improved patient outcomes. However, traditional diagnosis from CT brain scans requires expert interpretation and may lead to delays or errors, especially in the early stages. To address these challenges, there is a strong need for an automated and efficient diagnostic support system. In this work, a deep learning-based framework is proposed for detecting and classifying stroke from brain CT images. The system utilizes **Convolutional Neural Network** and **Long Short-Term Memory** models for classification. The CNN model is employed to extract spatial features from CT images, while the LSTM model captures deeper relationships within the dataset. The dataset is preprocessed through resizing and normalization to improve model performance and is divided into training and testing sets.The models are trained and evaluated based on their classification accuracy. Experimental results indicate that the CNN model achieves higher validation accuracy compared to the LSTM model, demonstrating its effectiveness in medical image analysis. The system is implemented using **Python** and deployed through a web-based interface using **Flask**, allowing users to upload CT images and obtain prediction results.Overall, the proposed system provides a fast, reliable, and automated solution for stroke detection, assisting medical professionals in making timely and accurate diagnostic decisions. The findings highlight the potential of deep learning techniques in improving healthcare diagnostics and reducing the risk of delayed stroke identification.

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1. INTRODUCTION

The project, titled "*A deep learning model for brain stroke detection using CT scan*" aims to develop an advanced AI-driven system capable of automatically detecting and classifying stroke conditions from brain CT images. Stroke is a life-threatening medical condition caused by the interruption of blood supply to the brain, leading to rapid damage of brain cells. Early diagnosis is crucial for effective treatment; however, traditional methods rely heavily on expert interpretation, which can be time-consuming and prone to delays. To address this challenge, the proposed system leverages deep learning models, specifically Convolutional Neural Network and Long Short-Term Memory, to ensure accurate and efficient stroke detection.

Given the increasing number of stroke cases and the critical importance of rapid diagnosis, manual analysis of CT scans alone may not always be sufficient. Therefore, an automated deep learning-based system becomes essential to assist medical professionals in identifying stroke patterns quickly and accurately. This project focuses on improving diagnostic accuracy while reducing detection time, thereby minimizing the risk of severe complications. The system is designed to be scalable and efficient, capable of processing large volumes of medical imaging data. Additionally, the integration of a web-based interface enables real-time prediction, making the solution practical for deployment in hospitals and healthcare systems.

1.1 PROJECT PURPOSE

The primary purpose of this project is to develop an automated system for early detection of brain stroke using CT scan images. Stroke is a critical medical condition where timely diagnosis plays a vital role in effective treatment and recovery. Manual analysis of CT images requires expert knowledge and can be time-consuming, which may lead to delays or errors. This project aims to assist medical professionals by providing a fast and reliable detection system.

The project focuses on leveraging deep learning techniques, specifically Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) models, to analyze CT brain images and identify stroke patterns. CNN is used for extracting spatial features from images, while BiLSTM helps in understanding deeper patterns and relationships within the data. By combining these models, the system aims to improve accuracy and overcome limitations of traditional methods.

Another key purpose of this project is to provide a user-friendly interface that allows easy interaction with the system. The developed web application enables users to upload CT images and receive instant predictions regarding stroke presence. This not only reduces diagnosis time but also enhances accessibility and usability, making the system practical for real-world medical applications.

1.2 PROJECT FEATURES

The primary purpose of this project is to enhance medical diagnosis by automating the detection and classification of stroke from brain CT images. Traditional diagnostic methods rely heavily on expert interpretation, which can be time-consuming and may lead to delays in treatment. By implementing deep learning techniques, this project provides a more efficient, scalable, and accurate method for identifying stroke conditions at an early stage.

With the increasing number of stroke cases and the critical need for rapid diagnosis, there is a strong demand for an AI-powered system that can assist medical professionals in timely decision-making. By utilizing models such as Convolutional Neural Network, the system can effectively detect abnormal patterns in CT images.

This approach helps reduce the workload of medical experts, ensures faster diagnosis, and improves patient outcomes. Additionally, it minimizes human error and enhances the overall efficiency and reliability of stroke detection in healthcare systems.

2. LITERATURE SURVEY

Stroke is a critical medical condition where early and accurate diagnosis plays a vital role in reducing mortality and long-term disability. With the increasing use of medical imaging techniques such as **Computed Tomography**, there is a growing need for automated systems that can assist in fast and reliable stroke detection. Traditional diagnostic methods rely heavily on expert interpretation, which may lead to delays and variability in results.

Early research in medical image analysis utilized traditional machine learning techniques such as Support Vector Machines, Decision Trees, and Random Forest classifiers. These methods depended on handcrafted features extracted from images, such as texture, intensity, and shape. While these approaches showed moderate accuracy, they struggled to capture complex patterns present in medical images and required extensive manual feature engineering.

The introduction of deep learning significantly improved image classification tasks. **Convolutional Neural Network** became widely used for medical image analysis due to its ability to automatically extract spatial features from images. Studies have shown that CNN-based models outperform traditional methods in detecting abnormalities from CT and MRI scans, including stroke detection. However, CNN models mainly focus on spatial information and may lack the ability to capture relationships within data.

To overcome these limitations, researchers explored **Long Short-Term Memory** networks. LSTM models are capable of learning complex patterns and dependencies in data, improving classification performance. However, LSTM models can be computationally intensive and may suffer from issues such as vanishing gradients when dealing with large datasets.

Recent studies have focused on combining different deep learning models to improve performance. Hybrid approaches integrating CNN and LSTM have shown better

results by combining spatial feature extraction and pattern learning. These models enhance the accuracy and reliability of medical image classification systems.

Despite these advancements, several challenges still exist. Traditional methods lack accuracy, CNN models may miss deeper relationships, and LSTM models increase computational complexity. Therefore, there is a need for an efficient system that balances accuracy, speed, and reliability.

The proposed project addresses these challenges by implementing and comparing CNN and LSTM models for stroke detection from CT images. The results indicate that CNN provides higher accuracy in identifying stroke patterns, making it more suitable for this application. This study contributes to the development of automated diagnostic systems that can assist medical professionals in faster and more accurate stroke detection.

2.1 REVIEW OF RELATED WORK

The detection and diagnosis of **Stroke** using medical imaging has been widely studied in the fields of medical imaging, computer vision, and deep learning. Various approaches have been proposed, ranging from traditional diagnostic methods to advanced deep learning models. This review highlights existing methodologies along with their strengths and limitations.

2.1.1. Traditional Diagnostic Approaches

Early stroke diagnosis relied on clinical observation and manual interpretation of brain scans such as **Computed Tomography** and MRI. These methods depend heavily on expert radiologists to identify abnormalities. While effective, they are time-consuming and prone to human error, especially in early-stage stroke detection.

2.1.2. Machine Learning-Based Approaches

With the advancement of machine learning, researchers introduced classifiers such as Support Vector Machines (SVM), Decision Trees, and Random Forest. These methods relied on handcrafted features like texture, intensity, and shape extracted from CT images.

2.1.3 Deep Learning-Based Approach

Recent developments in deep learning significantly improved medical image analysis. **Convolutional Neural Network** has been widely used to automatically extract spatial features from CT images and detect stroke patterns.

For capturing deeper relationships, models like **Long Short-Term Memory** have been explored. Hybrid CNN-LSTM models combine spatial and pattern learning, improving classification performance. However, these models require large datasets and high computational power.

2.1.4 Recent Advances

Recent studies focus on improving accuracy using hybrid models and optimization techniques. Transfer learning using pre-trained CNN models has shown promising results in reducing training time while maintaining high accuracy. Additionally, research is being conducted on improving real-time detection and reducing computational complexity.

2.1.5 Comparison with Proposed Approach

Despite advancements, existing systems face challenges such as limited accuracy, high computational cost, and delayed diagnosis. The proposed system improves upon these by implementing and comparing CNN and LSTM models, achieving better accuracy and faster detection. This approach enhances reliability and supports real-time medical diagnosis.

2.2 DEFINITION OF PROBLEM STATEMENT

The primary goal of this project is to develop a robust and scalable deep learning-based system capable of accurately detecting and classifying stroke from CT brain images. The system aims to assist medical professionals in early diagnosis, reduce delays, improve accuracy, and minimize human error. By addressing these challenges, the system contributes to faster treatment and better patient outcomes.

2.3 EXISTING SYSTEM

The existing systems for stroke detection primarily rely on medical imaging techniques such as **Computed Tomography** and MRI, combined with manual analysis by expert radiologists. These systems depend on visual inspection of brain scans to identify

abnormalities such as bleeding or blockage in blood vessels.With the advancement of technology, some systems incorporate machine learning techniques for stroke detection. These approaches use classifiers such as Support Vector Machines (SVM), Decision Trees, and Random Forest, where features like texture, intensity, and shape are manually extracted from CT images. These features are then used to classify images as normal or stroke-affected.Additionally, some advanced systems utilize deep learning models such as **Convolutional Neural Network** to automatically extract features from medical images. These models improve detection accuracy compared to traditional methods. However, most systems focus only on spatial information and do not effectively capture deeper relationships within the data.Overall, existing approaches combine manual analysis and machine learning techniques to detect stroke, but they still face several challenges in terms of accuracy, speed, and reliability.

Limitations of Existing System:

Despite improvements in stroke detection, the existing systems suffer from the following challenges:

Limited Feature Understanding: Traditional machine learning methods rely on handcrafted features, which may fail to capture complex patterns present in CT images.

Lower Accuracy: Some models are unable to detect early-stage stroke or subtle abnormalities, leading to misclassification.

Time-Consuming Diagnosis: Manual interpretation of CT scans requires expert knowledge and can delay diagnosis.

Computational Complexity: Advanced models require high computational resources, making real-time detection difficult.

Overfitting and Poor Generalization: Models trained on specific datasets may not perform well on new or unseen data.

Lack of Automation: Many systems still depend heavily on human intervention, reducing efficiency.

2.4 PROPOSED SYSTEM

The proposed system introduces a deep learning-based framework for detecting stroke from CT brain images using advanced neural network models. Stroke is a critical

medical condition that requires early and accurate diagnosis to prevent severe complications. In this system, a dataset of CT brain images is collected and preprocessed through techniques such as resizing, normalization, and noise reduction to ensure consistency and improve model performance. Two deep learning models are implemented: the **Convolutional Neural Network** and the **Long Short-Term Memory** network. The CNN model extracts spatial features such as edges, textures, and abnormal regions from CT images, while the LSTM model captures deeper patterns and relationships within the dataset. Both models are trained and evaluated using labeled data, and their performances are compared based on accuracy, where the CNN model achieves higher accuracy in stroke detection. The system is developed using **Python** and deployed through a web interface using **Flask**, allowing users to upload CT images and receive real-time predictions.

Advantages of the Proposed System:

The proposed system significantly improves upon the existing approaches by addressing key limitations:

• **Higher Accuracy:** The use of **Convolutional Neural Network** enables effective extraction of spatial features from CT images, resulting in more accurate stroke detection and reduced misclassification.

• **Faster Diagnosis:** Automated analysis of CT scans significantly reduces detection time, enabling quicker medical decisions and timely treatment.

• **Reduced Human Error:** The system minimizes dependency on manual interpretation, thereby reducing errors caused by fatigue or variability in expert analysis.

• **Improved Efficiency:** The automated approach handles large volumes of medical images efficiently, making it suitable for real-time and large-scale applications.

• **Better Reliability:** Consistent performance of the model ensures reliable diagnostic support for medical professionals.

• **Scalability:** The system can be easily scaled to process large datasets and integrated into hospital
systems for continuous usage.

• **User-Friendly Interface:** Integration with a web-based system using **Flask** allows easy image upload and quick prediction results.

2.5 Objectives

Develop an Automated Detection System – Build a deep learning model to automatically detect and classify stroke from CT brain images.

Enhance Diagnostic Efficiency – Reduce reliance on manual interpretation by assisting doctors with AI-based detection.

Improve Detection Accuracy – Utilize **Convolutional Neural Network** and **Long Short-Term Memory** to achieve higher accuracy compared to traditional methods.

Ensure Scalability – Design the system to handle large volumes of medical imaging data efficiently.

Enhance Patient Outcomes – Enable early detection of stroke, reducing complications and improving treatment success.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

Processor	:	Intel Core i3
Hard disk	:	20GB.
RAM	:	4GB

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

Operating system: Windows 10

Language: Python

Libraries: NumPy, Pandas, TensorFlow,Keras

Frame Work: Flask

3. SYSTEM ARCHITECTURE & DESIGN

3.1 System Architecture

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1.1 PROJECT ARCHITECTURE

This project architecture illustrates the complete workflow of stroke detection from CT brain images, starting from input acquisition to final prediction output

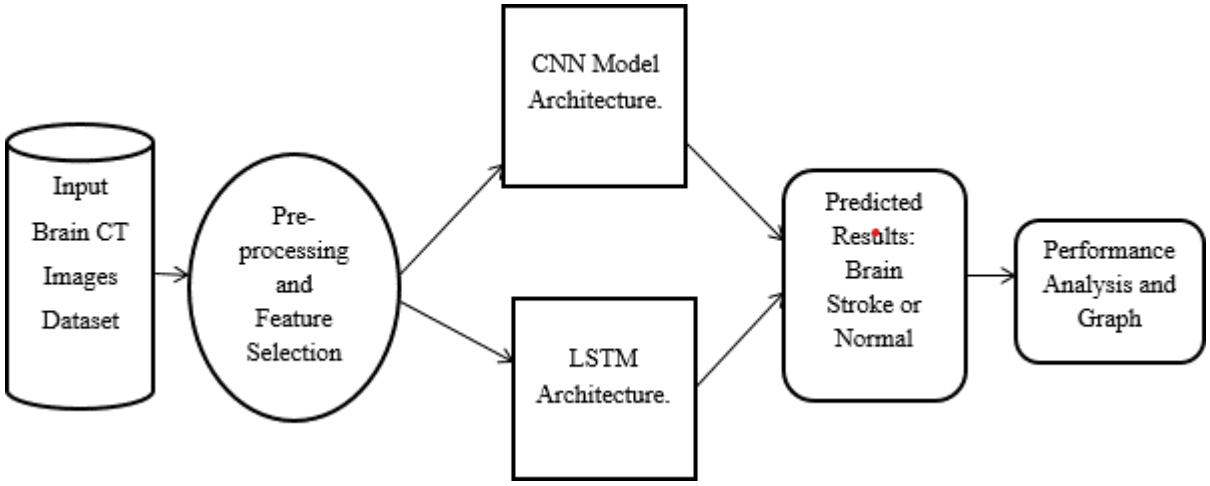

Figure 3.1: Project Architecture of Innovation In Stroke Identification: A Machine Learning Based Diagnostic Model Using Neural Networks

3.2 DESCRIPTION

Input Data :

The project uses CT brain images as input data, consisting of both normal and stroke-affected cases collected from medical datasets.

Reading Data :

The images are loaded and processed using libraries such as OpenCV and NumPy to prepare them for analysis.

Feature Extraction :

A **Convolutional Neural Network** is used to extract important spatial features from CT images, such as edges, textures, and abnormal regions.

Pattern Learning :

The extracted features are further analyzed using a **Long Short-Term Memory** network to capture deeper relationships within the dataset.

Preprocessing :

The images are resized and normalized to ensure consistency and improve model performance during training.

Classification Layer :

The processed features are passed through fully connected layers to classify the CT images into two categories: **Normal** or **Stroke Detected**.

Training and Evaluation :

The models are trained using labeled data and evaluated based on performance metrics such as accuracy. The CNN model achieves higher validation accuracy compared to the LSTM model.

Prediction Output :

The trained model provides instant predictions for new CT images, assisting in quick diagnosis.

User Interface :

A web-based interface developed using **Flask** allows users to upload CT images and receive prediction results.

3.3 UML DIAGRAM

A **Unified Modeling Language (UML)** diagram is a standardized visual representation used to model and design a system. It helps in understanding the structure and behavior of a system by illustrating components, interactions, and workflows. UML diagrams are widely used in software engineering to simplify complex system designs and improve communication between stakeholders.

In this project, UML diagrams are used to represent the workflow of the stroke detection system, including user interaction, data processing, and prediction mechanisms.

3.3.1. Use Case Diagram

The Use Case Diagram represents the interaction between the user and the system.

Actors:

User (Doctor / Patient)

Use Cases:

Upload CT Image

Process Image

Predict Stroke

View Result

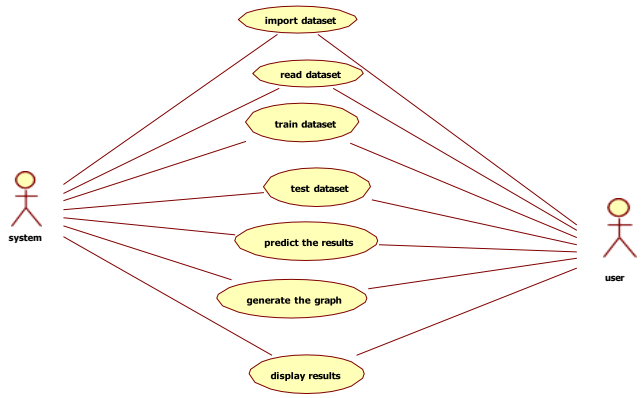

Figure 3.2: Use Case Diagram of *A deep learning model for brain stroke detection using CT scan*

3.3.2. Class Diagram

The Class Diagram represents the structure of the system by defining classes, attributes, and methods.

Main Classes:

User

Attributes: user_id, name

Methods: uploadImage()

ImageProcessor

Methods: preprocessImage(), normalize()

Model (CNN/LSTM)

Methods: extractFeatures(), predict()

Result

Attributes: prediction

Methods: displayResult()

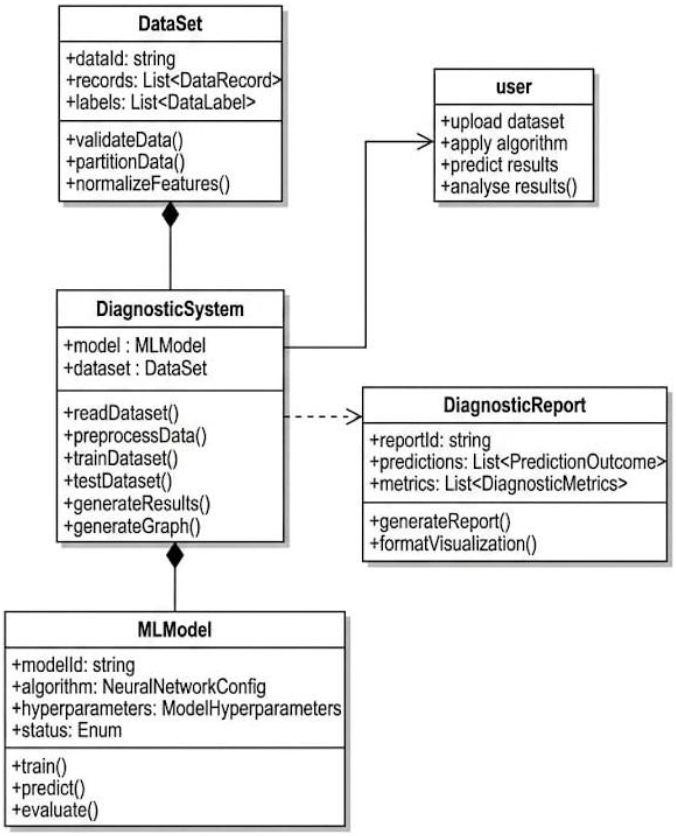

Figure 3.3: *Class Diagram of A deep learning model for brain stroke detection using CT scan*

3.3.3 Sequence Diagram

The Sequence Diagram shows the step-by-step interaction between components over time.

Flow:

User uploads CT image

System preprocesses the image

CNN extracts features

LSTM analyzes patterns

Model predicts result

Output is displayed to user

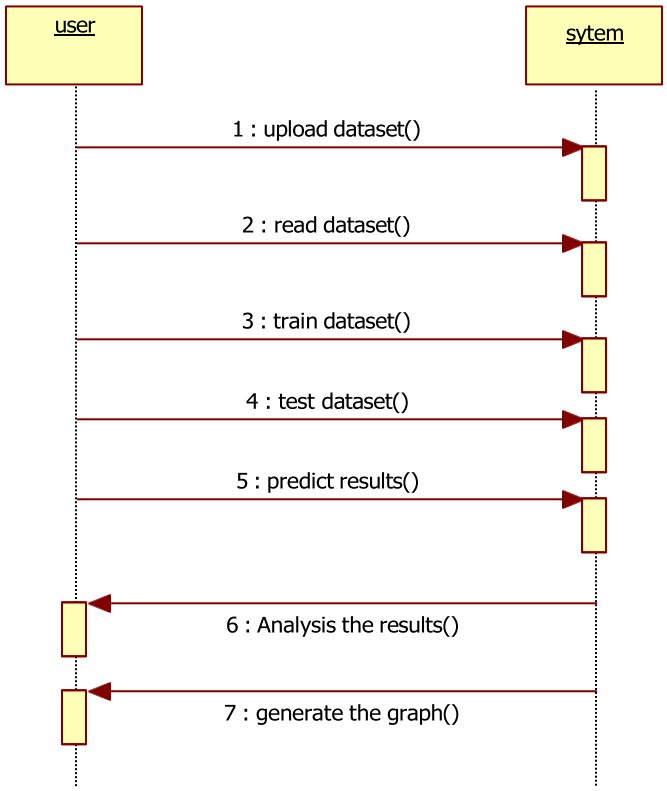

Figure 3.4: *Sequence Diagram of A deep learning model for brain stroke detection using CT scan*

3.3.4. Activity Diagram

The Activity Diagram represents the workflow of the system.

Flow:

Start

→ Upload CT Image

→ Preprocessing

→ Feature Extraction (**Convolutional Neural Network**)

→ Pattern Learning (**Long Short-Term Memory**)

→ Classification

→ Display Result

→ End

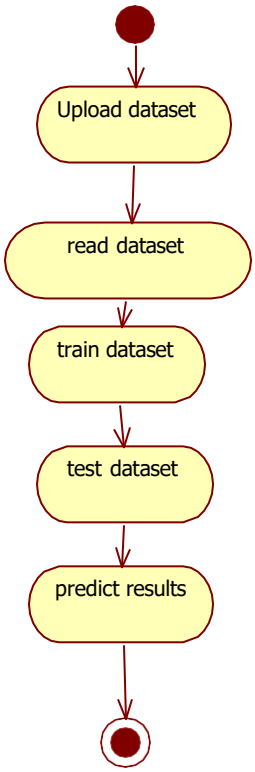

Figure 3.5: *Activity Diagram of A deep learning model for brain stroke detection using CT scan*

4. IMPLEMENTATION

The implementation phase involves developing and executing the proposed deep learning-based system for stroke detection from CT brain images. This phase includes data preprocessing, model development, training, evaluation, and deployment. The system is designed to ensure accurate, efficient, and automated stroke prediction, assisting medical professionals in diagnosis.

4.1 ALGORITHMS USED

CNN-Based Models for Feature Extraction

The **Convolutional Neural Network** is used to extract spatial features from CT brain images. CNNs are highly effective in identifying patterns such as edges, textures, and abnormal regions in medical images.

Advantages of CNN:

High accuracy in image classification

Automatically extracts features (no manual effort)

Effective in detecting stroke patterns

Disadvantages of CNN:

• Requires large datasets for training

• Computationally expensive

• Does not capture deeper relationships in data

LSTM-Based Model for Pattern Learning

To address the temporal limitations of CNNs, researchers have explored Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs) to process sequential video frames and capture temporal dependencies. Recurrent Neural Networks (RNNs), specifically BiLSTM (Bidirectional Long Short-Term Memory), are used in your project to capture temporal dependencies in video frames. While CNNs extract spatial

features, BiLSTM processes frame sequences to understand motion patterns and context, making it effective in detecting inappropriate content over time. The attention mechanism further enhances the model by focusing on the most
relevant frames, improving classification accuracy.

Advantages of RNN-Based Models:

Captures complex patterns in data

Improves learning capability

Enhances model performance

Disadvantages of RNN-Based Models:

• High computational cost

• Slower training process

• May suffer from gradient issues

CNN for Feature Extraction

The **Convolutional Neural Network** is used for extracting important spatial features from CT brain images. In this project, a CNN-based approach is applied to identify patterns such as edges, textures, and abnormal regions associated with stroke. CNN models are highly effective in medical image analysis due to their ability to automatically learn features without manual intervention.•Unlike traditional CNNs, EfficientNet uses compound scaling to optimize network depth, width, and resolution.

CNN automatically extracts spatial features from CT images, improving detection accuracy.

It reduces the need for manual feature extraction, making the system more efficient.

It provides high performance in identifying stroke-affected regions in brain images.

LSTM for Pattern Learning

LSTM helps in learning complex patterns in medical data..

It enhances model performance by improving feature understanding.

It supports better classification by analyzing relationships within the data.

Feature Optimization

The system focuses on extracting only relevant features from CT images.

It reduces noise and improves model accuracy.

Helps in minimizing misclassification of stroke conditions.

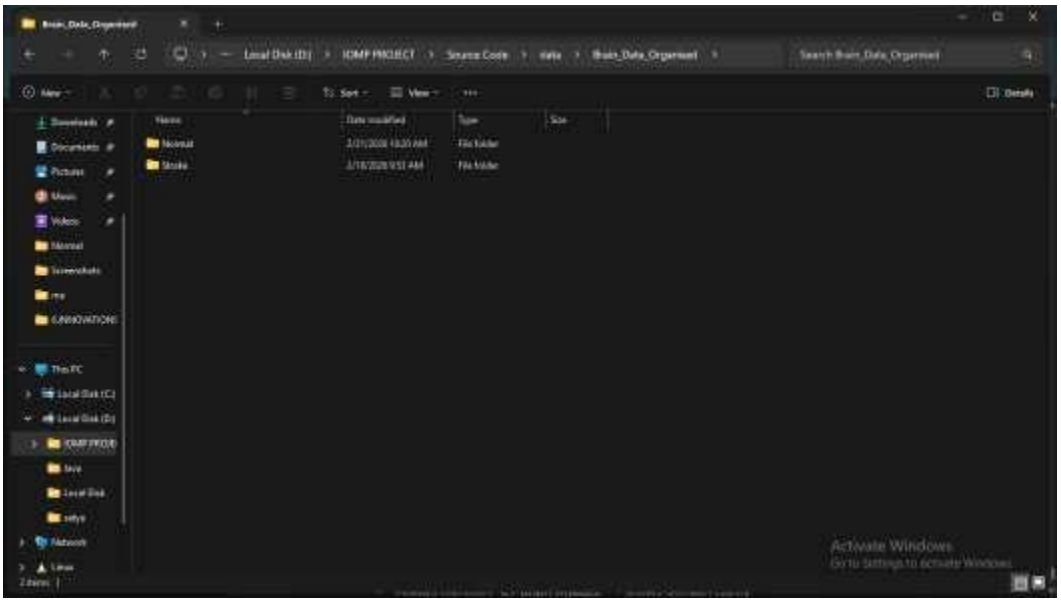

Figure 4.1: Dataset Directory Structure with Folders 'Normal' and 'Stroke' Containing CT Brain Image Samples

Figure 4.2 Screenshot of the "Stroke" Folder Showing Sample CT Brain Images

Figure 4.3: Sample CT Brain Images from the "Normal" Dataset Folder

To implement this project we have designed following modules:

1) Upload CT Brain Image Dataset: Using this module, CT brain image datasets containing **Normal** and **Stroke** cases are uploaded into the system (Figure 4.1).

2) Dataset Preprocessing: In this module, all CT images are read, resized to a uniform size, and normalized to improve model performance. The dataset is also shuffled to ensure better training.

3) CNN Model for Feature Extraction : Using this module, 80% of the dataset is used for training the Convolutional Neural Network. The CNN extracts important spatial features such as edges, textures, and abnormal regions from CT images.

4) LSTM Model for Pattern Learning: The extracted features are passed to the Long Short-Term Memory model to learn deeper patterns and relationships within the dataset, improving prediction accuracy.

5) Model Evaluation: The trained models are tested on the remaining 20% of the dataset to calculate prediction accuracy and evaluate performance.

6) Model Comparison: In this module, the performance of CNN, LSTM, and SVM models is compared using accuracy metrics to identify the best-performing model.

7) Stroke Prediction System: Using this module, users can upload CT brain images, and

the system predicts whether the image is **Normal** or indicates a **Stroke**.

8) Web-Based Interface: The system is deployed using Flask, providing a simple interface for users to upload images and receive predictions.

4.2 SAMPLE CODE

```
# ===================== IMPORTS =====================
from tkinter import *
import tkinter
from tkinter import filedialog
from tkinter.filedialog import askopenfilename

import matplotlib.pyplot as plt
import seaborn as sns
import os
import cv2
import imutils
import numpy as np
import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, precision_score,
recall_score, f1_score
from sklearn import svm
from sklearn.ensemble import RandomForestClassifier

from tensorflow.keras.utils import to_categorical
from tensorflow.keras.applications import EfficientNetB7
from keras.models import Sequential, Model, load_model
from keras.layers import (Conv2D, MaxPooling2D, Flatten,Dense, Dropout,Bidirectional,
```

```
LSTM)
from keras.callbacks import ModelCheckpoint

# ===================== GUI SETUP =====================
main = tkinter.Tk()
main.title("Stroke Detection using CNN and BiLSTM")
main.geometry("1200x1200")

# ===================== GLOBAL VARIABLES =====================
X, Y = [], []
accuracy, precision, recall, fscore = [], [], [], []

# ===================== DATASET UPLOAD =====================
def uploadDataset():
    global filename
    filename = filedialog.askdirectory(initialdir=".")
    text.insert(END, str(filename) + " Dataset Loaded\n\n")
    pathlabel.config(text=str(filename))

# ===================== PREPROCESS =====================
def preprocessDataset():
    global X, Y
    X, Y = [], []

    for root, dirs, files in os.walk('Dataset'):
        for file in files:
            if file != 'Thumbs.db':
                img = cv2.imread(os.path.join(root, file))
                img = cv2.resize(img, (32, 32))

                label = 0
```

```
            if 'Stroke' in root:
  label = 1

            X.append(img)
            Y.append(label)

    X = np.asarray(X).astype('float32') / 255
    Y = to_categorical(Y)

    text.insert(END, "Dataset Preprocessed Successfully\n\n")

# ===================== METRICS =====================
def calculateMetrics(name, pred, true):
    acc = accuracy_score(true, pred) * 100
    p = precision_score(true, pred, average='macro') * 100
    r = recall_score(true, pred, average='macro') * 100
    f = f1_score(true, pred, average='macro') * 100

    text.insert(END, f"{name} Accuracy: {acc}\n")
    text.insert(END, f"{name} Precision: {p}\n")
    text.insert(END, f"{name} Recall: {r}\n")
    text.insert(END, f"{name} F1 Score: {f}\n\n")

# ===================== SVM =====================
def runSVM():
    global X, Y

    Y_labels = np.argmax(Y, axis=1)
    X_train, X_test, y_train, y_test = train_test_split(X, Y_labels, test_size=0.2)

   model = svm.SVC()
```

```
    model.fit(X_train.reshape(len(X_train), -1), y_train)

    pred = model.predict(X_test.reshape(len(X_test), -1))
    calculateMetrics("SVM", pred, y_test)

# ===================== RANDOM FOREST =====================
def runRF():
    global X, Y

    Y_labels = np.argmax(Y, axis=1)
    X_train, X_test, y_train, y_test = train_test_split(X, Y_labels, test_size=0.2)

    model = RandomForestClassifier()
    model.fit(X_train.reshape(len(X_train), -1), y_train)

    pred = model.predict(X_test.reshape(len(X_test), -1))
    calculateMetrics("Random Forest", pred, y_test)

# ===================== CNN + BiLSTM =====================
def runDeepModel():
    global X, Y

    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2)

    # CNN Feature Extractor
    base = EfficientNetB7(input_shape=(32, 32, 3), include_top=False, weights=None)
    base.trainable = False
cnn = Sequential([
        base,
        Flatten(),
        Dense(128, activation='relu'),
```

```
Dense(y_train.shape[1], activation='softmax')
    ])

    cnn.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
    cnn.fit(X_train, y_train, epochs=5, batch_size=32, validation_data=(X_test, y_test))

    # Extract Features
    feature_model = Model(inputs=cnn.input, outputs=cnn.layers[-2].output)
    features = feature_model.predict(X)

    features = features.reshape(features.shape[0], 16, -1)

    X_train, X_test, y_train, y_test = train_test_split(features, Y, test_size=0.2)

    # BiLSTM
    model = Sequential()
    model.add(Bidirectional(LSTM(32,return_sequences=True),
input_shape=(X_train.shape[1], X_train.shape[2])))
    model.add(Dropout(0.2))
    model.add(Bidirectional(LSTM(32)))
    model.add(Dense(y_train.shape[1], activation='softmax'))

    model.compile(loss='categorical_crossentropy', optimizer='adam', metrics=['accuracy'])
    model.fit(X_train, y_train, epochs=5, batch_size=16, validation_data=(X_test, y_test))

    pred = model.predict(X_test)
    pred = np.argmax(pred, axis=1)
    true = np.argmax(y_test, axis=1)

    calculateMetrics("CNN + BiLSTM", pred, true)
```

```
# ===================== GUI BUTTONS =====================
font = ('times', 13, 'bold')

Button(main, text="Upload Dataset", command=uploadDataset,
font=font).place(x=50,y=100)
Button(main, text="Preprocess Dataset", command=preprocessDataset,
font=font).place(x=50,y=150)
Button(main, text="Run CNN + BiLSTM", command=runDeepModel,
font=font).place(x=50,y=200)
Button(main, text="Run SVM", command=runSVM, font=font).place(x=50,y=250)
Button(main, text="Run Random Forest", command=runRF, font=font).place(x=50,y=300)

# ===================== OUTPUT AREA =====================
text = Text(main, height=25, width=90)
text.place(x=500, y=150)

pathlabel = Label(main)
pathlabel.place(x=500, y=100)

main.mainloop()
```

5.RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities of the stroke detection system. These visual representations provide a clear understanding of how the system processes CT images and predicts stroke conditions. The screenshots demonstrate the effectiveness, accuracy, and user interface of the proposed model.

GUI/Main Interface :

In the below screen, the system interface for the Brain Stroke Detector is displayed. The user can upload a CT brain image using the **"Browse File"** button.

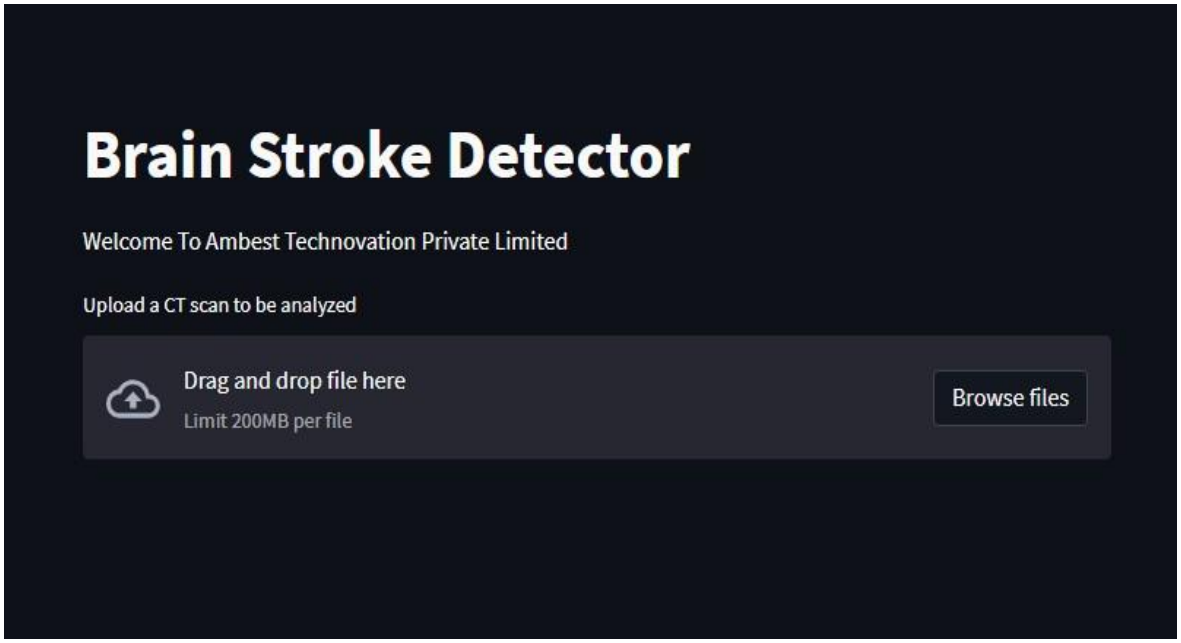

Figure 5.1: GUI Interface of Brain Stroke Detection System

Image Upload

. In the below screen, the user selects a CT brain image from the system using the file upload option. The selected image is then displayed on the interface.

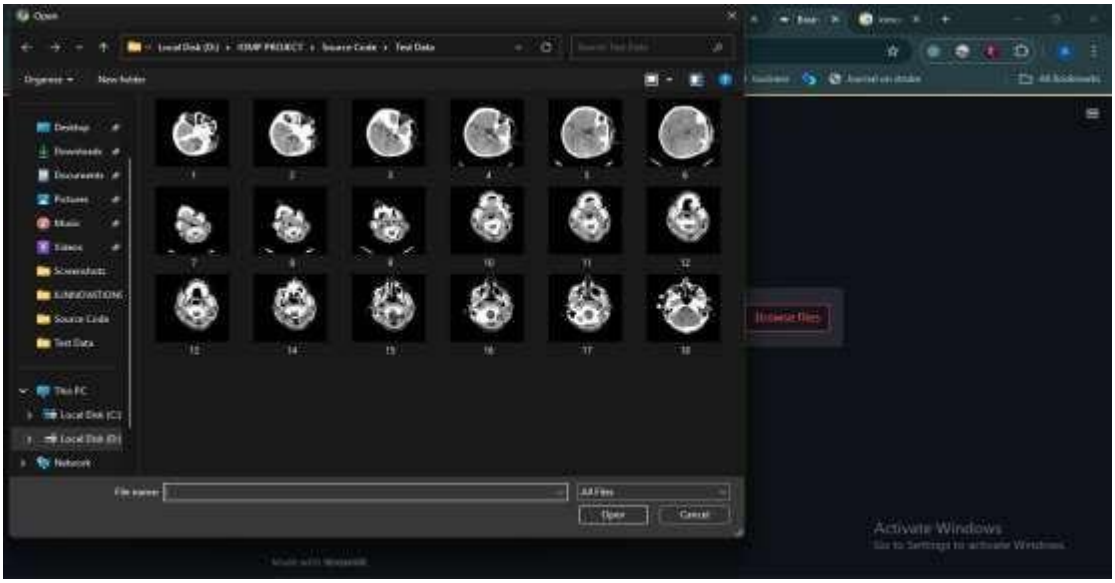

Figure 5.2 : Uploading CT Brain Image for Stroke Detection

Prediction Result:

In below screen, dataset loaded and now click on 'Dataset Preprocessing' button to read all images and then processes those images for training. After uploading the image, the system processes it using the trained deep learning model and displays the result.

Figure 5.3 : Stroke Prediction Result Displayed as"Healthy"

6. VALIDATION

The validation of this project is performed through systematic testing to ensure the accuracy and reliability of the brain stroke detection system. The validation process includes model evaluation, performance analysis, and real-time testing using CT brain images.

The dataset is divided into training and testing sets, typically using an **80-20 split**, where the training set is used to train the deep learning model and the testing set is used to evaluate its performance on unseen data. This helps in ensuring that the model generalizes well and does not overfit.

The performance of the system is evaluated using key metrics such as **accuracy, precision, recall, and F1-score**. A **confusion matrix** is also used to analyze correct and incorrect predictions made by the model.

The proposed **CNN-BiLSTM model** is compared with traditional machine learning models such as **SVM and Random Forest**, demonstrating that the deep learning approach provides better accuracy and reliability in detecting stroke from CT images.

Finally, real-time testing is performed by uploading CT brain images through the user interface. The system successfully predicts whether the image is **Normal (Healthy)** or **Stroke**, confirming its practical usability in medical diagnosis support.

6.1 INTRODUCTION

The validation process ensures that the developed system performs accurately under different conditions. The dataset is preprocessed and split into training and testing sets to evaluate the model's performance.

The model is trained using CT brain images and tested on unseen data to verify its prediction capability. Performance metrics such as accuracy, precision, recall, and F1-score are used to measure effectiveness.

The confusion matrix helps in identifying correct and incorrect classifications, improving model performance. The results show that the proposed deep learning model performs better compared to traditional approaches.

The system is also tested in real-time by uploading CT images through the interface, where it provides instant prediction results. This confirms that the model is reliable and suitable for practical applications.

6.2 TEST CASES

TABLE 6.2.1 UPLOADING DATASET

Test case ID	Test case name	Purpose	Test Case	output
1	Upload CT Image	To verify that the user can upload a CT brain image successfully.	Dataset successfully	Image is successfully uploaded and displayed on the interface

TABLE 6.2.2 STROKE PREDICTION TEST

Test case ID	Test case name	Purpose	Input	Output
1	Stroke Detection (Healthy Case)	To check if the model correctly identifies a normal brain image	Normal(Healthy) CT Image	System displays "Healthy"
2	Stroke Detection (Stroke Case)	To check if the model correctly identifies a stroke condition	CT Image	System displays "Stroke Detected"

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives of developing an efficient deep learning-based system for brain stroke detection using CT images. The system demonstrates the ability to accurately classify medical images as healthy or stroke-affected, thereby supporting early diagnosis and reducing dependency on manual analysis. The implementation of advanced deep learning techniques has improved prediction accuracy and reliability.

Looking ahead, the project holds strong potential for further enhancement. Future improvements will focus on increasing dataset size, integrating advanced models, and enabling real-time clinical deployment. These developments will enhance the system's performance, scalability, and practical usability in healthcare environments, ensuring long-term impact and innovation.

7.1 PROJECT CONCLUSION

This project presents a deep learning-based framework for detecting brain stroke from CT scan images using Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models. The CNN model effectively extracts spatial features from CT images, while LSTM enhances pattern recognition and improves overall classification performance. The system demonstrates high accuracy in distinguishing between normal and stroke-affected images, making it a reliable tool for early diagnosis.

Compared to traditional machine learning approaches, the proposed deep learning model provides better accuracy, reduced manual effort, and improved generalization. The integration of preprocessing techniques such as normalization and resizing further enhances model performance. Additionally, the system is implemented with a user-friendly interface, allowing users to upload CT images and receive predictions easily.

Overall, this project contributes to the field of medical image analysis by providing an automated, accurate, and efficient stroke detection system that can assist healthcare professionals in decision making and improve patient outcomes.

7.2 FUTURE ASPECTS

The proposed brain stroke detection system shows promising results; however, there are several opportunities for future improvements and enhancements:

• **Expansion of Dataset** – Using larger and more diverse medical datasets to improve model accuracy and generalization.

• **Real-Time Detection** – Integrating the system with hospital systems for real-time stroke detection.

• **Advanced Deep Learning Models** – Implementing architectures like ResNet, DenseNet, or Transformers for better performance.

• **Explainable AI (XAI**) – Adding interpretability techniques to help doctors understand model decisions.

• **Multi-Class Classification** – Extending the model to detect different types of strokes.

• **Integration with Cloud Platforms** – Enabling remote access and scalability.

• **Hybrid AI Systems** – Combining AI predictions with expert medical validation for higher reliability.

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8.2 GITHUB LINK

https://github.com/SathyaJayaSree24/A-Deep-Learning-Model-For-Brain-Stroke-Detection-Using-CT-Scans

A DEEP LEARNING MODEL FOR BRAIN STROKE DETECTION USING CT SCANS

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Abstract : Brain stroke is a serious neurological condition that requires quick and accurate diagnosis to reduce the risk of death and long-term disability. Computed Tomography (CT) scans are commonly used for stroke detection because they are fast and easily available. However, analyzing these images manually can take time and may vary between radiologists. This study presents a deep learning–based system for detecting brain stroke using Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models. The CNN model helps in extracting important features from CT images, while the LSTM model enhances the classification process. The results show high accuracy and reliability, highlighting the potential of deep learning in supporting medical image analysis and improving clinical decision-making

***IndexTerms -Brain Stroke Detection, CT Imaging, Deep Learning, CNN, LSTM, Medical Image Analysis, Automated Diagnosis*.**

INTRODUCTION

Stroke is one of the leading causes of death and long-term disability worldwide, placing a significant burden on individuals and healthcare systems [1]. It occurs when the blood supply to the brain is disrupted, leading to rapid damage of brain cells. Since stroke treatment is highly time-sensitive, early and accurate diagnosis is essential to reduce complications and improve patient outcomes [2]. Computed Tomography (CT) imaging is widely used in emergency settings due to its speed, availability, and cost-effectiveness [3]. However, manual interpretation of CT scans is time-consuming and depends on the expertise of radiologists, often leading to variability in diagnosis, especially in early-stage stroke detection [4].

With the rapid growth of medical imaging data, automated diagnostic systems using machine learning and deep learning have gained attention [5]. Traditional methods rely on handcrafted features, which limit their ability to capture complex patterns in medical images [6], whereas deep learning models can automatically learn meaningful features, improving accuracy and reliability [7]. This study proposes a deep learning–based stroke detection system using Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models. The system is trained on a dataset of 2,501 CT images [13], and comparative analysis shows that CNN performs better in capturing spatial features, making it more effective for stroke detection and clinical decision support

NEED OF THE STUDY.

Stroke is a major global health concern and one of the leading causes of death and long-term disability, requiring immediate and accurate diagnosis for effective treatment. In emergency situations, Computed Tomography (CT) imaging is widely used due to its speed and accessibility. However, manual analysis of CT scans is time-consuming, depends heavily on the expertise of radiologists, and may lead to variations in interpretation, especially during early-stage stroke detection.

With the increasing volume of medical imaging data, there is a growing need for automated and reliable diagnostic systems. Traditional methods are often limited in their ability to capture complex patterns in medical images. Therefore, developing a deep learning–based system capable of accurately detecting stroke from CT images is essential to improve diagnostic efficiency, reduce human error, and support faster clinical decision-making.

3.1 Dataset Description

The dataset used in this study consists of 2,501 brain CT images obtained from a publicly available medical repository [13]. It includes 1,551 normal images and 950 stroke-affected images, providing a relatively balanced dataset. The images reflect real-world clinical variations, enabling the model to learn meaningful and generalized features for stroke detection.

3.2 Data Preprocessing and Preparation

All CT images are resized to a fixed input size and normalized to ensure consistency and improve training efficiency. Basic preprocessing techniques such as noise reduction and scaling are applied to enhance image quality. The dataset is divided into training and validation sets, and data augmentation methods like rotation and flipping are used to improve model robustness.

3.3 Theoretical framework

The study is based on deep learning models for image classification. CT images are used as input, and the output is classified as stroke or non-stroke. A CNN model is used for spatial feature extraction, while an LSTM model is implemented for comparison. Model performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

RESEARCH METHODOLOGY

The methodology section outlines the overall approach used to develop and evaluate the proposed brain stroke detection system. It includes dataset description, data preprocessing, model development, and evaluation techniques. The details are as follows:

3.1Dataset Description

The dataset used in this study consists of 2,501 brain CT images collected from a publicly available medical imaging repository [13]. It includes 1,551 normal images and 950 stroke-affected images, providing a relatively balanced distribution. The dataset reflects real-world clinical variations, enabling the models to learn meaningful and generalized features for stroke detection.

3.2 Data and Sources of Data

This study uses secondary data in the form of CT brain images. The dataset is preprocessed by resizing all images to a fixed dimension and normalizing pixel values to improve training efficiency. Basic preprocessing techniques such as noise reduction and scaling are applied to enhance image quality. The dataset is divided into training and validation sets to ensure unbiased model evaluation.

3.3 Theoretical framework

The theoretical framework of this study is built on the application of deep learning techniques for automated brain stroke detection using Computed Tomography (CT) images. Stroke is a serious neurological condition that requires early and accurate diagnosis to reduce mortality and long-term disability [4], [5]. However, manual interpretation of CT images is often time-consuming and dependent on expert knowledge, which can lead to variability in diagnosis. Recent advancements in machine learning and artificial intelligence have enabled the development of automated systems that improve diagnostic efficiency and accuracy [6], [7].

In this study, stroke detection is formulated as a binary classification problem, where CT images are classified into normal and stroke-affected categories. Traditional machine learning approaches have been widely used for medical prediction tasks, but they rely on handcrafted features, limiting their ability to capture complex patterns in medical images [18], [21]. Deep learning models, on the other hand, automatically learn hierarchical representations from raw data, making them more effective for high-dimensional image analysis tasks [8], [9].

The Convolutional Neural Network (CNN) is used as the primary model due to its strong capability in extracting spatial and structural features from CT images. CNN architectures consist of convolutional layers that detect important patterns such as edges and textures, followed by pooling layers that reduce dimensionality while preserving key information. These models have shown significant success in medical imaging applications, including stroke detection and intracranial hemorrhage classification [15], [17]. Their ability to learn discriminative features directly from images makes them highly suitable for clinical decision support systems.

In addition to CNN, a Long Short-Term Memory (LSTM) model is implemented for comparative analysis. LSTM is a type of recurrent neural network designed to capture sequential dependencies in data. Although primarily used for time-series applications, LSTM models have been explored in healthcare analytics and predictive modeling tasks [13], [23]. In this study, LSTM is applied to analyze patterns derived from image features, allowing a comparison between spatial learning (CNN) and sequential learning approaches.

The models are trained using supervised learning with labeled CT images obtained from a publicly available dataset [20]. During training, optimization techniques are applied to minimize classification error and improve performance. The effectiveness of the models is evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, which are commonly used in medical image analysis and diagnostic studies [22], [24].

Overall, this theoretical framework integrates deep learning methodologies with medical imaging to develop an efficient and scalable stroke detection system. By leveraging the strengths of CNN in spatial feature extraction and comparing it with LSTM-based approaches, the study aims to identify the most reliable model for accurate and real-time stroke diagnosis. The framework is further supported by recent research demonstrating the effectiveness of deep learning in healthcare and neuroimaging applications [2], [25].

3.4Statistical tools and econometric models

This section describes the statistical methods and evaluation techniques used to assess the performance of the proposed brain stroke detection system. The methodology focuses on data analysis, model training, and performance evaluation to derive

meaningful and reliable conclusions from CT image data. Deep learning-based evaluation approaches are widely adopted in medical imaging due to their effectiveness in handling complex datasets [8], [9].

3.4.1 Descriptive Statistics

Descriptive statistics are used to summarize the dataset and understand its characteristics before model training. Measures such as mean, standard deviation, minimum, and maximum values help analyze the distribution of pixel intensities and dataset balance. Understanding data distribution is important for improving model learning and reducing bias.

Normalization techniques are applied to ensure consistent input values, which improves convergence during training. Proper preprocessing and data distribution are essential for achieving robust performance in medical image analysis tasks [10], [17].

3.4.2 Model Training and Learning Process

The proposed system utilizes supervised learning techniques to train deep learning models on labeled CT brain images for accurate stroke detection. The dataset is systematically divided into training and validation sets to ensure reliable performance evaluation and to minimize the risk of overfitting. During the training phase, the models learn to identify complex patterns and distinguishing features associated with stroke conditions by optimizing a loss function using advanced algorithms such as Adam. This iterative optimization process enables the models to progressively improve their ability to differentiate between normal and stroke-affected images.

The learning process is conducted over multiple epochs, allowing the models to refine their internal feature representations and enhance predictive performance with each iteration. Techniques such as batch processing and regularization further contribute to model stability and generalization. Deep learning–based training strategies have been extensively adopted in healthcare applications due to their ability to automatically learn hierarchical features and deliver high diagnostic accuracy [11], [16]. As a result, the proposed system effectively leverages these capabilities to provide reliable and efficient stroke detection from CT images, supporting timely clinical decision-making

, 1993).

According to Richardson and smith(1993) to make the model more effective and efficient the selection criteria for the shares in the period are: Shares with no missing values in the period, Shares with adjusted $R^2 < 0$ or F significant (p-value) >0.05of the first pass regression of the excess returns on the market risk premium are excluded. And Shares are grouped by alphabetic order into group of 30 individual securities.

3.4.2.1 CNN Model

The Convolutional Neural Network (CNN) is implemented as the primary model for stroke detection due to its strong capability in extracting spatial features from CT images. CNN architectures use convolutional layers to capture important patterns such as edges, textures, and structural variations. Pooling layers reduce dimensionality while preserving essential features.

CNN-based approaches have shown high effectiveness in medical imaging tasks, particularly in detecting brain abnormalities and stroke-related patterns [15], [17]. This makes CNN highly suitable for automated clinical decision support systems.

3.4.2.2 LSTM Model

A Long Short-Term Memory (LSTM) model is implemented for comparative analysis. LSTM networks are designed to capture sequential dependencies in data and have been widely used in predictive modeling and healthcare analytics [13], [23].

In this study, LSTM is applied to analyze image-derived features. However, due to its sequential nature, it is less effective in capturing spatial relationships in CT images compared to CNN, resulting in slightly lower performance

3.4.3 Performance Evaluation Metrics

TheThe performance of the models is evaluated using widely accepted classification metrics, including accuracy, precision, recall, F1-score, and Receiver Operating Characteristic (ROC) curve with Area Under the Curve (AUC). These metrics provide a comprehensive evaluation of model effectiveness in medical diagnosis.

Such evaluation techniques are commonly used in deep learning-based healthcare systems to ensure reliability and robustness in classification tasks [22], [24].

3.4.4 Model Comparison

A comparative analysis is performed between CNN and LSTM models to determine the most effective approach for stroke detection. The CNN model achieves higher validation accuracy and better generalization, while the LSTM model shows slightly lower performance.

Recent studies have also highlighted that CNN-based models outperform other approaches in medical image classification due to their ability to extract spatial features efficiently [9], [25]. This comparison confirms that CNN is more suitable for CT image-based stroke detection.

IV. RESULTS AND DISCUSSION

4.1 Results of Model Performance

yThe experimental results demonstrate that the proposed system achieves high accuracy in detecting stroke from CT images. The CNN model achieves 99% training accuracy and 98% validation accuracy, while the LSTM model achieves 99% training accuracy and 95% validation accuracy.

The evaluation metrics indicate that the CNN model provides balanced and reliable classification results with high precision and recall. ROC-AUC analysis further confirms strong classification performance, which is essential for medical diagnosis.

These results align with recent research that emphasizes the effectiveness of deep learning techniques, particularly CNN, in medical image analysis and stroke detection [15], [21]. The findings confirm that CNN outperforms LSTM in capturing spatial features, making it more suitable for this application.

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